Zach Mobille

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SUMMARY

Ph.D. candidate in Quantitative Biosciences with 7+ years of experience applying sophisticated statistical and predictive analytics — including machine learning, data mining, and inferential analysis — to high-dimensional structured and unstructured data. Proven track record of end-to-end ownership of data science solutions, from design through technical delivery and measurable outcome. Proficient in Python; experienced collaborating with cross-functional teams and communicating actionable insights to diverse stakeholders. Seeking to leverage quantitative modeling expertise and emerging methods in a data science role focused on minimizing risk and improving business outcomes.

EDUCATION

Georgia Institute of Technology Atlanta, GA
Ph.D. in Quantitative Biosciences (GPA 3.9/4.0) Expected Graduation: Aug. 2026
Concentration in Computational Neuroscience

- **Dissertation**: Information Coding and Structural Motifs in Spiking Neural Networks
- **Academic Awards**: ARCS Foundation Award Scholarship, InQuBATE training fellowship

Illinois State University Bloomington, IL
Master of Science in Biomathematics (GPA 3.8/4.0) May 2021

- **Thesis**: A Computational Investigation of the Biophysical Mechanisms Underlying Thermotaxis in the AFD Neurons of *Caenorhabditis elegans*

Illinois State University Bloomington, IL
Bachelor of Science in Physics (GPA 3.6/4.0) May 2019
Minor in Mathematics

- **Capstone project**: Analog Implementation of the Hodgkin-Huxley model neuron
- **Academic Awards:** Dean's List, Redbird Academic Scholarship

TECHNICAL SKILLS

- **Programming languages**: Python, C/C++, MATLAB, Julia, Java, R, SAS, Mathematica, Bash
- **Data Tools**: Scikit-learn, PyTorch, Pandas, Keras, SciPy, NumPy
- **High-Performance Computing**: Slurm workload manager
- **Quantitative Techniques**: machine learning, predictive modeling, inferential statistical analysis, data mining, data visualization, time series analysis, risk analytics

SELECTED PROJECTS

Turning Earnings Call Data into Early Warnings for Securities Fraud 2026
Georgia Tech Hacklytics competition

- Implemented LightGBM model to predict the likelihood of fraud at a company, based on prior earnings call data.
- Applied data preprocessing and feature engineering to structured text and audio data, aligning financial features to quarterly fraud records to reduce business risk.
- Achieved PR-AUC of 0.091 for predicted fraud within a 4-year time horizon, 5X random baseline of 0.018.

Option Price Data Prediction with Black-Scholes Model
Independent project 2025

- Utilized the Black-Scholes quantitative risk model to predict closing prices of call options from public NVIDIA market data, demonstrating domain application of financial analytics
- Predicted 200 option prices with 93% accuracy using current stock price, implied volatility, option strike price

Jane Street Real-Time Market Data Forecasting Kaggle Competition
Independent project 2025

- Performed data mining and implemented predictive models (linear regression, SVMs, LSTMs, GRUs) on a 79-dimensional set of anonymized financial returns, encompassing both structured and unstructured temporal data.
- Applied data preprocessing by imputing missing values for models that do not accept them.
- Applied cross-validation with Bayesian optimization to ensure model robustness and generalization; exercised sound judgment in model development to guard against overfitting and performance risk.
- Achieved >60% prediction accuracy on test data, as measured by the coefficient of variation R^2.

Mushroom Classification with Machine Learning Bloomington, IL
MAT 443 – Machine Learning & Data Mining course Aug. – Dec. 2020

- Conducted end-to-end data mining project: applied inferential statistical analysis on categorical features and evaluated a suite of predictive models (decision trees, logistic regression, random forests, neural networks) for binary outcome classification, achieving 100% accuracy with multiple methods.

RESEARCH EXPERIENCE

Georgia Institute of Technology Atlanta, GA
Graduate Research Assistant, Mathematical Neuroscience Lab Jan. 2021 - Present

- Apply sophisticated statistical and predictive analytics to high-dimensional data, extracting actionable insights and communicating findings to diverse scientific audiences through publications and conference presentations.
- Collaborate with cross-functional teams across mathematics, neuroscience, and biology to align end-to-end data science solutions with domain requirements, fostering strategic partnerships with stakeholders across disciplines.
- Publish 2 peer-reviewed papers in leading journals and present at conferences, actively advancing emerging methods in computational data science applied to neural systems.
- Emphasize reusability and scalability in analysis pipelines by implementing consistent coding standards and best practices, with rigorous documentation to ensure reproducibility of all computational results.
- Act as a resource for junior researchers, providing mentorship and technical guidance on quantitative methods and computational approaches in neuroscience.

Illinois State University Bloomington, IL
Graduate Research Assistant, Computational Neuroscience Lab Aug. 2019 – May 2021

- Built and validated mathematical models of electrochemical dynamics in neurons underlying temperature-tracking behavior in *C. elegans*.
- Communicated with experimentalists to align computational modeling approach with biological realism.
- Published a first-author research paper and presented findings at several seminars and conferences.

Illinois State University Bloomington, IL
Undergraduate Research Assistant, Computational Neuroscience Lab Aug. 2017 – May 2019

- Developed a quantitative model of astrocyte signaling in a pair of coupled neurons, implemented in C++.
- Performed data visualization and manuscript revisions for a paper where synchronization of neurons was studied.
- Designed and constructed an electronic circuit that models a biological neuron.
- Mentored a junior student in electronics and co-authored 2 papers published in leading journals.

LEADERSHIP & COMMUNICATION EXPERIENCE

Georgia Institute of Technology, Atlanta, GA Aug. 2021 - Present
Graduate Teaching & Lecture Assistant, Grader

- Courses taught: Introductory Physics, Probability and Statistics with Applications
- Courses graded: Dynamics & Bifurcation 1, Computational Neuroscience
- Evaluated and provided detailed feedback on assignments, exams, and worksheets.
- Mentored students during weekly recitation sessions and office hours on complex quantitative topics.